

Exploring Latent Brain Networks via Simultaneous Component Analysis

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Sections 4–5 summarize results of Helwig & Snodgrass (2019)

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Color, Music, and Emotion (Whiteford et al., 2018)

Suppose you want to study music-to-color associations.

- Music-color synesthesia: hearing music $\longrightarrow$ seeing color
- Robust and systematic associations exist in nonsynesthetes

Emotional mediation hypothesis: colors and music are related indirectly via higher-level emotional associations.

Three-Way Data: subjects rate colors and music on emotional scales.

- Subjects $\times$ Emotions $\times$ Stimuli
- How should we analyze these data???

Some Analytical Options

Assume latent dimensions account for common variation in ratings, so consider some application of Factor Analysis (FA).

Option 1: FA of concatenated data ($\text{Subjects} \times \text{Emotions} * \text{Stimuli}$)

Option 2: FA of data averaged across subjects ($\text{Stimuli} \times \text{Emotions}$)

Option 3: Separate FA for each stimuli ($\text{Subjects} \times \text{Emotions}$)

None of these options is ideal for understanding the ratings data:

1. conflates variation in emotions and stimuli
2. ignores individual differences in ratings
3. difficult to connect disparate solutions

Parallel Factor Analysis: Emotions and Subjects

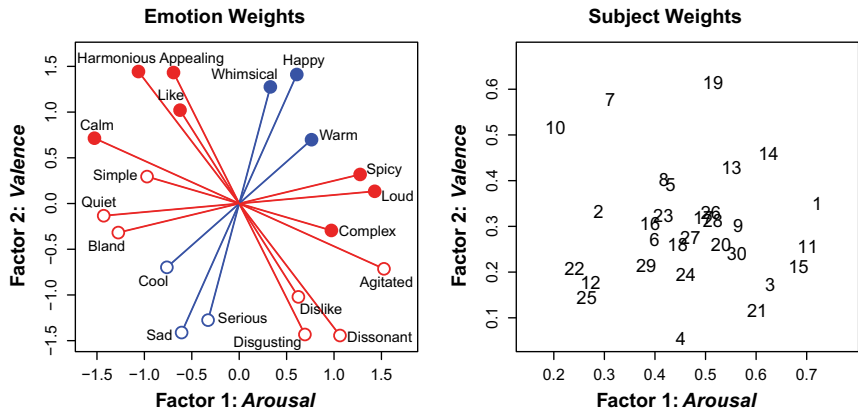


Figure 1: Emotion and Subject weights.

Parallel Factor Analysis: Colors and Music

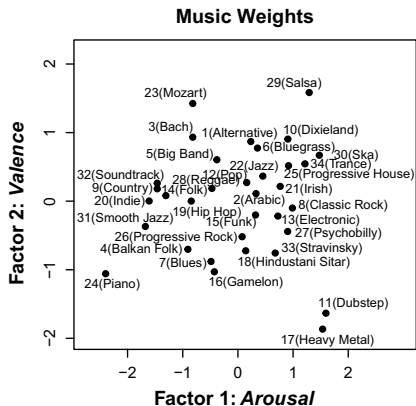
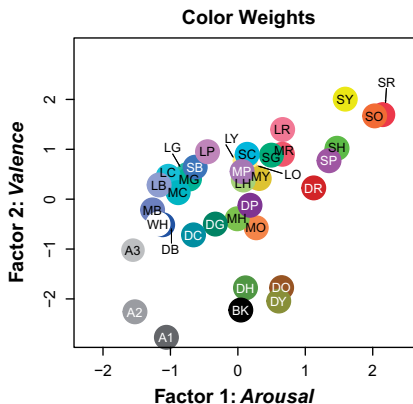


Figure 2: Color and Music weights.

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Bilinear Model (FA and PCA and ICA)

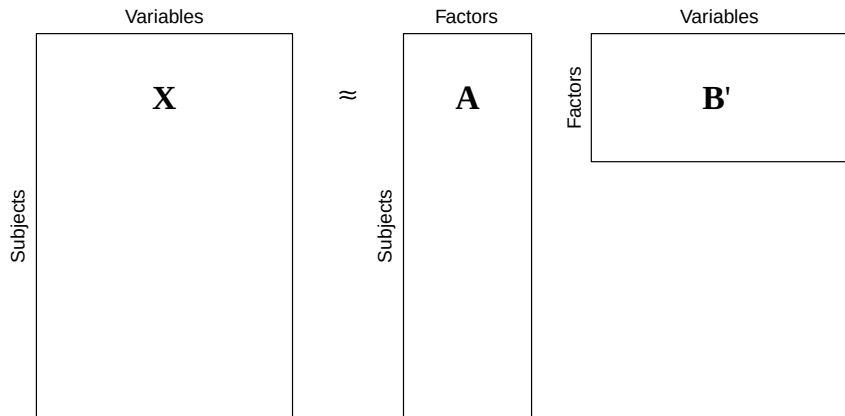


Figure 3: Bilinear model (Pearson, 1901; Hotelling, 1933)

Parallel Factors (Parafac) Model

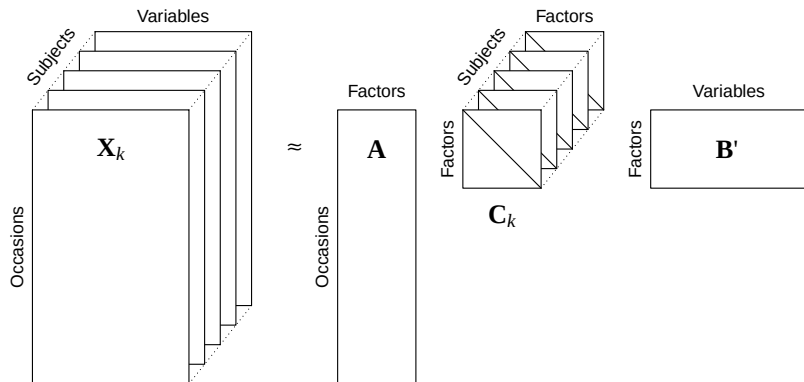


Figure 4: Parafac model (Hitchcock, 1927; Harshman, 1970; Carroll & Chang, 1970)

Parallel Factors-2 (Parafac2) Model

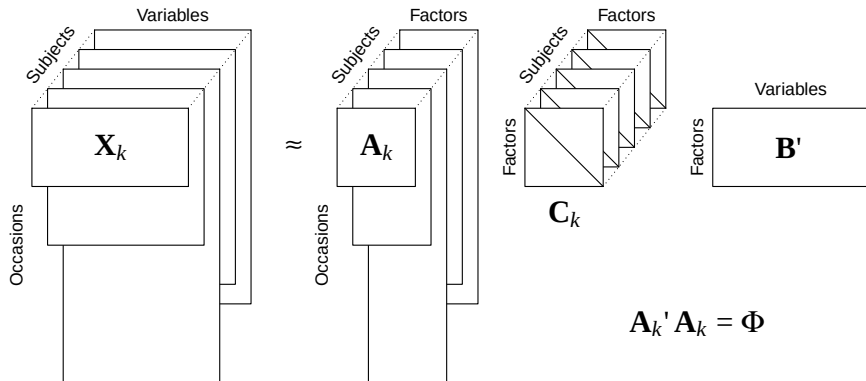


Figure 5: Parafac2 model (Harshman, 1972)

Simultaneous Component Analysis (SCA) Models

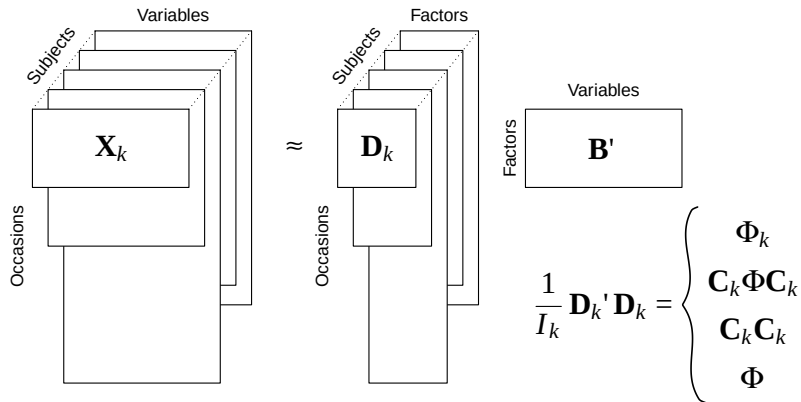


Figure 6: SCA models (Timmerman & Kiers, 2003), which are known as SCA-P (Φ_k), SCA-PF2 ($\mathbf{C}_k \Phi \mathbf{C}_k$), SCA-IND ($\mathbf{C}_k \mathbf{C}_k$), and SCA-ECP (Φ)

Benefits of Multiway Component Models

Leverages multiway structure to analyze variation across all modes

- No need to concatenate or average data
- No need to connect (disparate) slice-wise results

Some models (e.g., Parafac and Parafac2) produce unique solutions

- No rotational indeterminacy or factor score indeterminacy
- Intrinsic axis property lets data determine factor orientation

Easily extendible to higher order data arrays (e.g., 4-way or 5-way)

Primer on Neuroimaging Studies

Event-related¹ studies record brain signals in response to some event.

- Goal is to understand neural processes elicited by event

Typical studies collect multi-subject spatiotemporal data:

- time $\times$ space $\times$ subjects
- time $\times$ space $\times$ subjects $\times$ conditions
- time $\times$ space $\times$ subjects $\times$ conditions $\times$ trials

To make sense of data, some dimension reduction approach is needed.

¹Known as “task-based” studies for fMRI data

Independent Components Analysis

Neuroimage studies often rely on ICA for dimension reduction

- ICA = PCA with rotation to maximize statistical independence

Multiway/tensor extensions of ICA have been proposed

- Tensor ICA (Beckmann & Smith, 2005) uses Parafac model form
- Group ICA (Beckmann et al., 2005) uses SCA-P model form

Problem: Are latent brain signals statistically independent???

- Assumption is questionable (Helwig & Hong, 2013)

Solution: Parafac2 can uniquely find latent brain networks

Depiction of Neuroimage Component Models

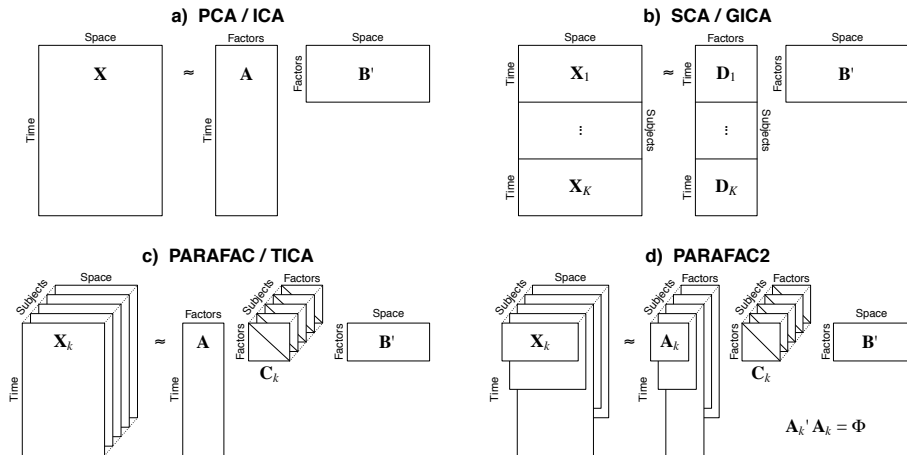


Figure 7: Component models for neuroimage data (time $\times$ space $\times$ subjects).

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Overview of Analysis Procedure

Fitting a multiway component model requires several decisions:

1. How should data be pre-processed?
2. Which multiway model should be used?
3. How many factors should be extracted?
4. Should constraints be applied to weights?
5. How many random starts should be used?

Some of these have familiar analogues in bilinear models.

- Ex: choosing the number of factors

Other considerations are unique to multiway models.

- Ex: choosing the number of random starts

Multiway Data Pre-Processing

Pre-processing refers to how the data are centered and/or scaled.

- Centering removes constants across levels of mode(s)
- Scaling equalizes influence within levels of mode(s)

For FA and PCA, there are conventions for data pre-processing:

- Center data across subjects for each variable
- Scale variables to have equal variance

For multiway models, there are no pre-processing conventions

- Can center and/or scale any mode of data
- Depends on the covariance you want to capture

Constraints in Multiway Models

Constraints refer to what we assume (or require) of the weights.

- Default is to apply no constraints to any mode's weights
- Some possibilities: orthogonality, non-negativity, etc.

In FA and PCA, it is customary to assume the scores are orthogonal

- Rotational indeterminacy $\iff$ orthogonality inconsequential
- Other constraints (e.g., non-negativity) are rarely considered

For Parafac and Parafac2, constraints fundamentally change solution

- Intrinsic axis property implies you can't "rotate away" constraints
- Constraints are useful for stabilizing estimation with noisy data

Alternating Least Squares Estimation

Weights are typically estimated via alternating least squares (ALS):

1. Initialize all weight matrices
2. Update each weight matrix given others
3. Repeat Step 2 until convergence

Converges to a locally optimal solution, which depends on Step 1

- Try many random starts to find global optimum
- Prevalence of local optima depends on fit model

Step 2 uses unconstrained or constrained least squares for each update.

Fitting Multiway Models in R

multiway (Helwig, 2019) is an R package for fitting multiway models

- Constraints implemented via **CMLS** package (Helwig, 2018a)

Fit models include:

- Canonical Polyadic Decomposition (`cpd`)
- Individual Differences Scaling (`indscal`)
- Multiway Covariates Regression (`mcr`)
- Parallel Factor Analysis 1 (`parafac`)
- Parallel Factor Analysis 2 (`parafac2`)
- Simultaneous Component Analysis (`sca`)
- Tucker Factor Analysis (`tucker`)

24 Constraint Options for `cmls()` Function

Table 1: Constraint options for `cmls()` function.

Label	M	N	O	P	S	U	Details
uncons							unconstrained
nonneg		✓					non-negative
period				✓			periodic
pernon		✓		✓			periodic and non-negative
smooth					✓		smooth
smonon		✓			✓		smooth and non-negative
smoper				✓	✓		smooth and periodic
smpeno		✓		✓	✓		smooth, periodic, and non-negative
orthog			✓				orthogonal
ortnon		✓	✓				orthogonal and non-negative
ortsmo			✓		✓		orthogonal and smooth
orsmpe			✓	✓	✓		orthogonal, smooth, and periodic
moninc	✓						monotonic increasing
monnon	✓	✓					monotonic increasing and non-negative
monsmo	✓				✓		monotonic increasing and smooth
mosmno	✓	✓			✓		monotonic increasing, smooth, and non-negative
unimod						✓	unimodal
uninon		✓				✓	unimodal and non-negative
uniper				✓		✓	unimodal and periodic
unpeno		✓		✓		✓	unimodal, periodic, and non-negative
unismo					✓	✓	unimodal and smooth
unsmno		✓			✓	✓	unimodal, smooth, and non-negative
unsmpe				✓	✓	✓	unimodal, smooth, and periodic
unsmpn		✓		✓	✓	✓	unimodal, smooth, periodic, and non-negative

Arguments for `parafac()` Function

```
parafac(X, nfac, nstart = 10, const = NULL, control = NULL,  
        Afixed = NULL, Bfixed = NULL, Cfixed = NULL, Dfixed = NULL,  
        Astart = NULL, Bstart = NULL, Cstart = NULL, Dstart = NULL,  
        Astruc = NULL, Bstruc = NULL, Cstruc = NULL, Dstruc = NULL,  
        Amodes = NULL, Bmodes = NULL, Cmodes = NULL, Dmodes = NULL,  
        maxit = 500, ctol = 1e-4, parallel = FALSE, cl = NULL,  
        output = c("best", "all"), verbose = TRUE, backfit = FALSE)
```

- Specify and control the constraints
- Provide fixed weights for a given mode
- Provide starting values for a given mode
- Provide structural constraints for a given mode
- Provide mode search ranges for unimodality

Pre- and Post-Processing Functions

multiway package has some functions for pre-processing data

- `ncenter()` centers multiway data
- `nscale()` scales multiway data

multiway package has some functions for post-processing results

- `reorder()` reorders the factors of a fit model
- `resign()` resigns a specified mode's weights
- `rescale()` rescales a specified mode's weights

multiway package has some other useful functions

- `fitted()` constructs the model implied (fitted) tensor
- `congru()` for Tucker's congruence coefficient (Tucker, 1951)
- `corcondia()` for core consistency diagnostic (Bro & Kiers, 2003)

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Simulation Design: Overview

Simulate realistic EEG data from visual stimulus ERP study

- $I = 256$ time points $\times J = 61$ electrodes $\times K = 30$ subjects
- $R = 2$ components motivated by real data results (next slide)

Fully crossed simulation design with 24 cells

- 4 levels of time course heterogeneity: $\sigma \in \{0, 0.5, 1, 1.5\}$
- 6 levels of signal to noise ratio: $\text{SNR} \in \{0.1, 0.15, 0.25, 0.4, 0.6, 1\}$

To introduce realistic heterogeneity, define $\mathbf{A}_k = \mathbf{F}\boldsymbol{\alpha}_k$ where

- $\mathbf{F} = \{f_\ell(i)\}_{I \times \nu}$ is a B-spline basis function matrix with $\nu = 30$
- $\boldsymbol{\alpha}_k = [\boldsymbol{\alpha}_{k,1}, \boldsymbol{\alpha}_{k,2}]$ with $\boldsymbol{\alpha}_{k,r} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_r, \sigma^2 \boldsymbol{\Sigma}_r)$

Simulation Design: Time Courses and Spatial Maps

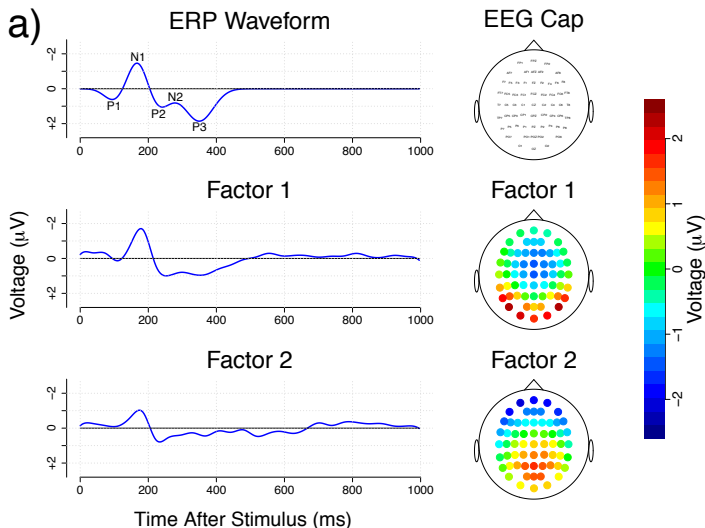


Figure 8: ERP time courses and spatial maps. Factor 1 is the Ventral processing stream and Factor 2 is the Dorsal processing stream.

Simulation Design: Time Course Heterogeneity

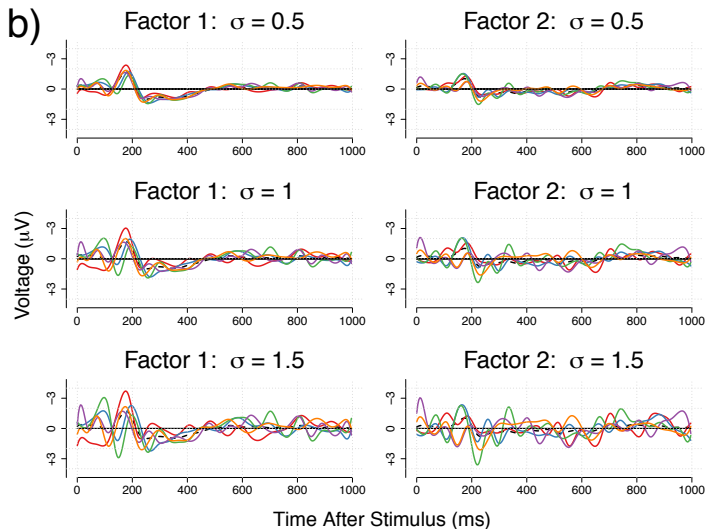


Figure 9: Examples of time course heterogeneity. When $\sigma = 0$, data follow Parafac model. When $\sigma > 0$, data follow SCA model. Using $\sigma = 1$ is representative of real data.

Simulation Analyses: Data Generation and Methods

For each of the 24 cells, generate 100 replications of the tensor

- $\mathbf{A}_k = \mathbf{F}\boldsymbol{\alpha}_k$ where $\boldsymbol{\alpha}_k = [\boldsymbol{\alpha}_{k,1}, \boldsymbol{\alpha}_{k,2}]$ with $\boldsymbol{\alpha}_{k,r} \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}_r, \sigma^2 \boldsymbol{\Sigma}_r)$
- $\mathbf{B}$ is fixed at spatial maps in Figure 8
- $\mathbf{C}$ is generated independently from Uniform[0.5, 1.5]

For each data replication, fit six different models:

- Parafac with NN constraint on mode C
- Parafac2 with NN constraint on mode C (see Helwig, 2013)
- SCA-P with Varimax rotation (Kaiser, 1958)
- SCA-P with no rotation
- GICA with FastICA rotation (Hyvärinen, 1999)
- GICA with FastICA and dual-regression (Beckmann et al., 2009)

Simulation Analyses: Performance Metrics

To assess performance, models were fit using $R = 2$ and $\nu = 30$

Performance was assessed using a variety of different metrics:

- Tensor recovery:
 - R-squared for observed tensor
 - Root mean squared error (RMSE) for latent tensor
- Factor recovery:
 - Tucker congruence coefficient for time courses (A)
 - Tucker congruence coefficient for spatial maps (B)

RMSE: smaller values are better. Others: values closer to 1 are better.

Simulation Analyses: K-Fold Cross-Validation

CV-CORCONDIA to select the number of factors R

- Fit model using $R = 1, \dots, 4$
- Thresholds of > 80 , > 85 , > 90
- Compare to optimal solution with $R = 2$

CV-MSE to select the degrees of freedom ν (smoothness)

- Fit model using $\nu \in \{10, 15, \dots, 75, 80\}$
- Threshold of $\delta = 0.2$ standardized MSE change
- Compare to optimal solution with $\nu = 30$ and unsmooth ($\nu = 256$)

Simulation Results: Tensor Recovery

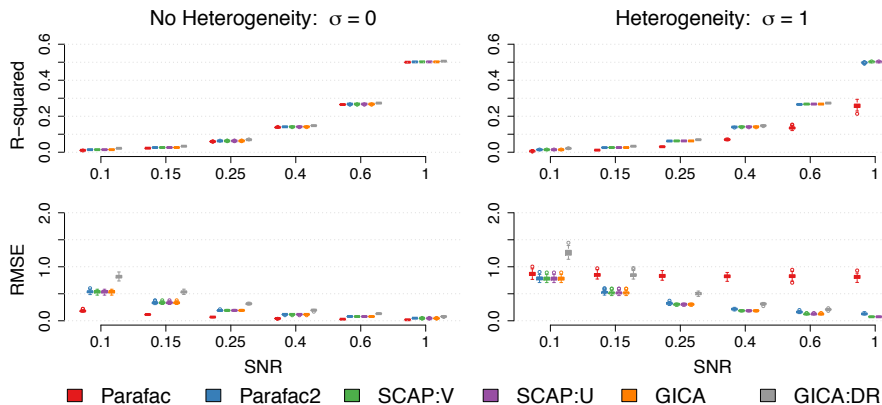


Figure 10: Performance metrics for tensor recovery.

Simulation Results: Factor Recovery

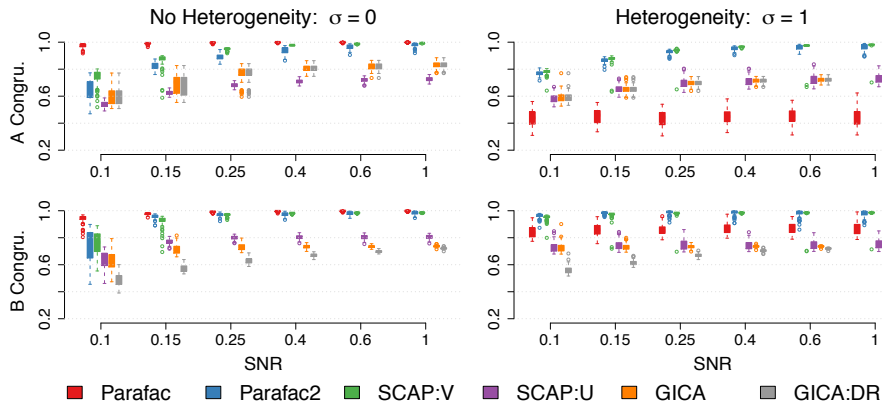


Figure 11: Performance metrics for factor recovery.

Simulation Results: K-Fold Cross-Validation (Part 1)

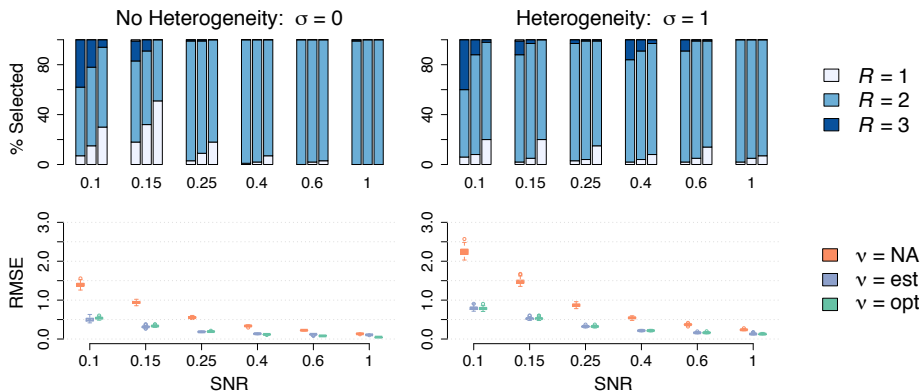


Figure 12: K-fold CV results for factor and smoothness selection.

Simulation Results: K-Fold Cross-Validation (Part 2)

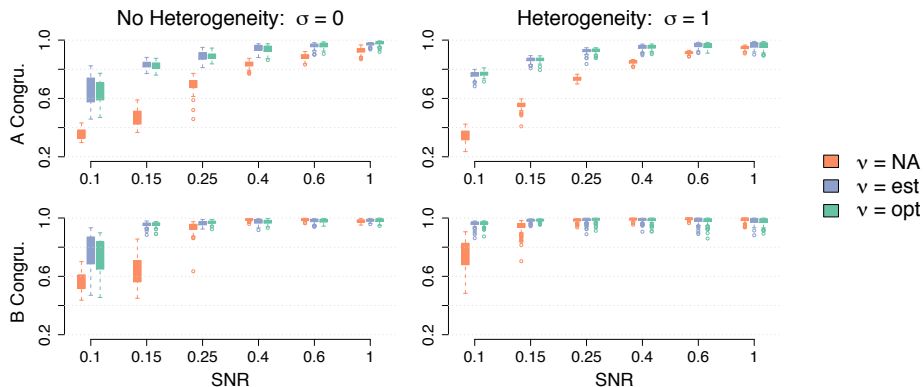


Figure 13: K-fold CV results for smoothness selection.

Simulation Results: Key Findings

Tensor recovery results reveal that...

- R-squared isn't too helpful for choosing correct model
- RMSE of Parafac2 and SCA-P can be very similar

Factor recovery results reveal that...

- Parafac2 recovers factors—even when SCA-P is correct model
- SCA-P and GICA may not (b/c of rotational indeterminacy)

K-fold cross validation results reveal that...

- CV-CORCONDIA can be useful for choosing number of factors
- CV-MSE can be useful for choosing degree of smoothness

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EEG Data: Overview

EEG data from UCI machine learning repository (Dua & Graff, 2019)

- Henri Begleiter Neurodynamics Laboratory at SUNY Downstate
- Compiled and posted by Ingber (1997, 1998)

61-channel visual stimulus ERPs recorded at 256 Hz for one second

- Visual stimuli from the Snodgrass & Vanderwart (1980) stimuli set
- Used data from the single stimulus condition (S1)

122 subjects (77 alcoholics and 45 controls) in the full dataset

- Restrict analyses to 115 subjects (73 alcoholics and 42 controls)
- Required 20+ trials with artifact free data in S1 condition

EEG Data: Electrode Cap

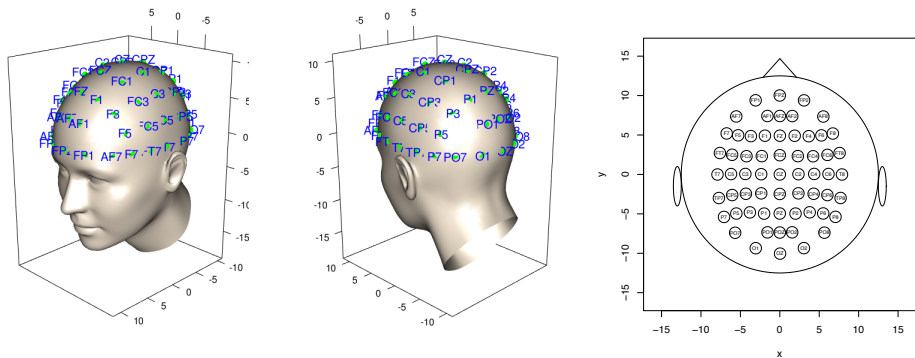


Figure 14: Visualization of 61-channel EEG cap from `eegkit` R package (Helwig, 2018b).

EEG Data: Preprocessing

Artifact: voltage magnitude greater than $100 \mu V$ ($\sim 5\%$ of trials)

- Typically due to movement or signal drift (due to bad connection)

Bandpass filtered (1-50 Hz) and then averaged across first 20 trials

- Removes electrical signals that are too low or too high

Re-referenced (from Cz) to an average reference

- Voltages are relative to average at each time point

Centered across time for each subject and electrode

- Ensures models analyze the covariance across time

EEG Analyses

Preprocessing resulted in a three-way tensor of EEG data

- 256 time points $\times$ 61 electrodes $\times$ 115 subjects

Step 1: Use k-fold cross validation to choose R and ν

- Tune CV-CORCONDIA with respect to $R = 1, \dots, 4$
- Tune CV-MSE with respect to $\nu = 10, 15, \dots, 75, 80$

Step 2: Fit multiway models using chosen R and ν

- Fit same six models used in simulation study
- Used same fitting specifications for fitting

EEG Results: K-Fold Cross Validation

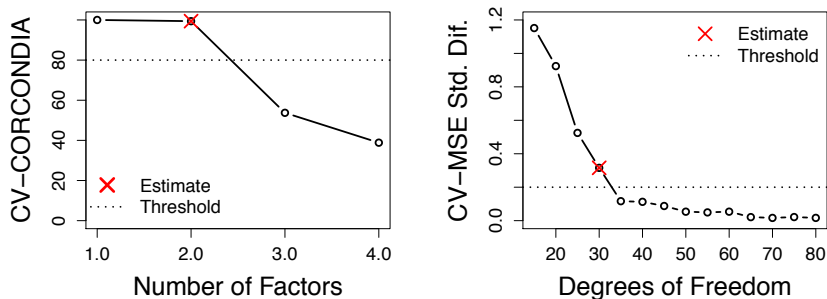


Figure 15: K-Fold cross validation for R and ν .

EEG Results: Explained Variation

Using the chosen dimensionality $R = 2$ and smoothness $\nu = 30$

Method	R-squared
Parafac	0.2981
Parafac2	0.6334
SCA-P	0.6411
GICA	0.6411
GICA-DR	0.6939

R-squared results reveal that...

- Parafac is not flexible enough to capture heterogeneity
- Parafac2 produces similar fit as SCA-P and GICA
- GICA-DR squeezes out another 5-6% variance accounted for

EEG Results: Spatial Maps

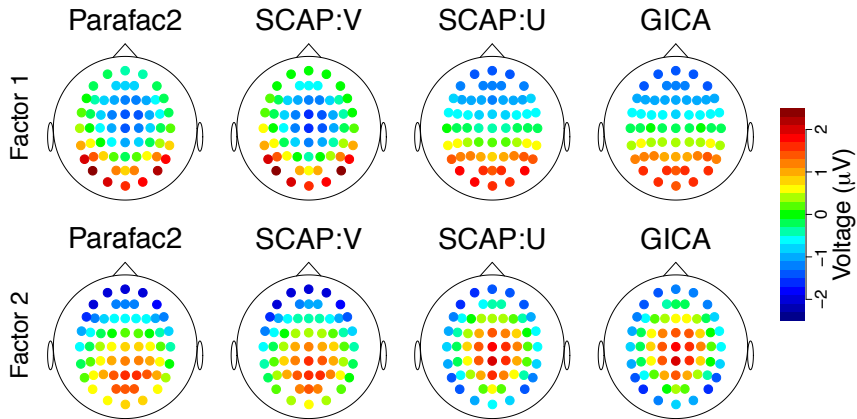


Figure 16: Estimated spatial maps for each method.

EEG Results: Parafac2 Solution

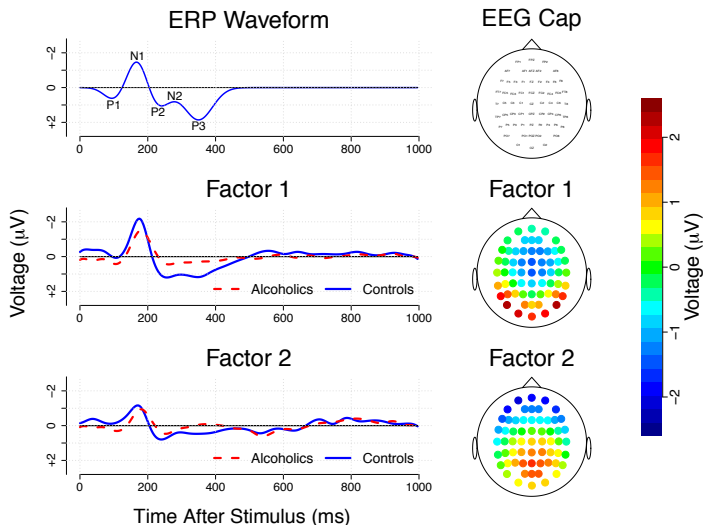


Figure 17: Parafac2 time courses and spatial maps. Time courses show average mode A weights for alcoholics and controls.

EEG Results: Ventral versus Dorsal Stream

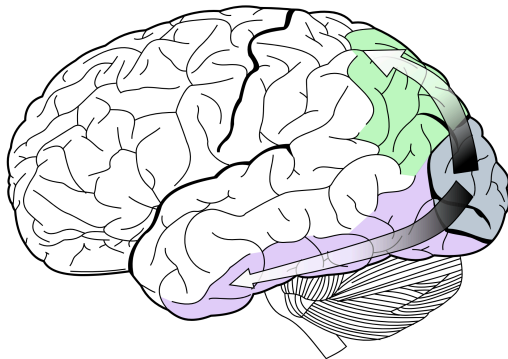


Figure 18: Dorsal stream (green) and ventral stream (purple). Dorsal stream is “where” (or “vision-for-action” path) and ventral stream is “what” (or “vision-for-perception” path). Figure from https://en.wikipedia.org/wiki/Two-streams_hypothesis

EEG Results: Alcoholics versus Controls

R-squared differences between alcoholics and controls

- 59.6% for alcoholics (30.5% dorsal and 29.1% ventral)
- 67.0% for controls (39.3% for dorsal and 27.7% ventral)

Alcoholics show attenuated ERP voltages in both streams

- Peaks and valleys of average ERP waveforms are attenuated
- Supports past findings about reduced P300 in alcoholics

Alcoholics show noteworthy delays in key ERP events

- Delay in P1 peak for dorsal stream and N1 peak for both streams
- Suggests information processing abnormality in early visual stream

EEG Results: Key Findings

K-fold cross validation results reveal that...

- $R = 2$ and $\nu = 30$ work well for these data
- K-fold CV can be effective with real data

Model comparison results reveal that...

- Parafac2 fit data about as well as more flexible models
- Parafac2 produces a unique interpretation of results

Alcoholic versus control results reveal that...

- Alcoholics have more heterogeneity in their ERPs
- Alcoholics have visual information processing abnormalities

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Take Home Message

Multiway models have immense potential for psychological data

- Applicable whenever variables are recorded multiple times
- Longitudinal multivariate data are becoming common (e.g., EMA)

Multiway models can find the “true” latent dimensions

- No need to rely on (subjective) post-hoc factor rotations
- Intrinsic axis property lets the data do the talking

Cross validation can be useful for tuning multiway models

- Can be used to choose model dimensionality
- Can be used to choose model constraints

Room for Improvement

Choosing between multiway models can be difficult

- Competing models can have similar fit statistics
- No clear way of choosing which model is best

Choosing number of factors can be difficult

- Proposed CV-CORCONDIA works better with higher SNR
- Hard to reliably find factors that explain small R-squared

Conducting statistical inference can be difficult

- Bootstrap is possible if one mode is randomly sampled
- Special care is needed when bootstrapping (Kiers, 2004)

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Appendix Table of Contents

1. Alternating Least Squares Algorithms
2. Functional Parafac and Parafac2
3. Constrained Multivariate Least Squares
4. CMLS Simulation

Parafac ALS Algorithm² (Harshman, 1970)

Define the matricized versions of the tensor:

- $\mathbf{X}_A = [\mathbf{X}_1, \dots, \mathbf{X}_K]$ ($I \times JK$ matrix)
- $\mathbf{X}_B = [\mathbf{X}'_1, \dots, \mathbf{X}'_K]$ ($J \times IK$ matrix)
- $\mathbf{X}_C = [\text{vec}(\mathbf{X}_1), \dots, \text{vec}(\mathbf{X}_K)]'$ ($K \times IJ$ matrix)

ALS Algorithm:

0. Initialize $\mathbf{B} = \{b_{jr}\}_{J \times R}$ and $\mathbf{C} = \{c_{kr}\}_{K \times R}$
1. $\mathbf{A} = \mathbf{X}_A \mathbf{Z}_A (\mathbf{Z}'_A \mathbf{Z}_A)^{-1}$ where $\mathbf{Z}_A = (\mathbf{C} \odot \mathbf{B})$
2. $\mathbf{B} = \mathbf{X}_B \mathbf{Z}_B (\mathbf{Z}'_B \mathbf{Z}_B)^{-1}$ where $\mathbf{Z}_B = (\mathbf{C} \odot \mathbf{A})$
3. $\mathbf{C} = \mathbf{X}_C \mathbf{Z}_C (\mathbf{Z}'_C \mathbf{Z}_C)^{-1}$ where $\mathbf{Z}_C = (\mathbf{B} \odot \mathbf{A})$

Iterate steps 1–3 until convergence of the SSE loss function

²Actually due to Robert I. Jennrich at UCLA.

Parafac2 ALS Algorithm (Kiers et al., 1999)

$$\mathbf{A}_k' \mathbf{A}_k = \Phi \iff \mathbf{A}_k = \mathbf{P}_k \mathbf{H} \quad \text{where} \quad \mathbf{P}_k' \mathbf{P}_k = \mathbf{I}_R \quad \text{and} \quad \mathbf{H}' \mathbf{H} = \Phi$$

ALS Algorithm:

0. Initialize $\mathbf{H} = \{h_{sr}\}_{R \times R}$ and $\mathbf{B} = \{b_{jr}\}_{J \times R}$ and $\mathbf{C} = \{c_{kr}\}_{K \times R}$
1. Update mode A weights
 - (a) $\mathbf{P}_k = \mathbf{U}_k \mathbf{V}_k'$ where $\mathbf{U}_k \mathbf{S}_k \mathbf{V}_k'$ denotes the SVD of $\mathbf{X}_k \mathbf{B} \mathbf{C}_k' \mathbf{H}'$
 - (b) $\mathbf{H} = \tilde{\mathbf{X}}_A \mathbf{Z}_A (\mathbf{Z}_A' \mathbf{Z}_A)^{-1}$ where $\tilde{\mathbf{X}}_A = [\tilde{\mathbf{X}}_1, \dots, \tilde{\mathbf{X}}_K]$ with $\tilde{\mathbf{X}}_k = \mathbf{P}_k' \mathbf{X}_k$
2. $\mathbf{B} = \tilde{\mathbf{X}}_B \tilde{\mathbf{Z}}_B (\tilde{\mathbf{Z}}_B' \tilde{\mathbf{Z}}_B)^{-1}$ where $\tilde{\mathbf{Z}}_B = (\mathbf{C} \odot \mathbf{H})$
3. $\mathbf{C} = \tilde{\mathbf{X}}_C \tilde{\mathbf{Z}}_C (\tilde{\mathbf{Z}}_C' \tilde{\mathbf{Z}}_C)^{-1}$ where $\tilde{\mathbf{Z}}_C = (\mathbf{B} \odot \mathbf{H})$

Iterate steps 1–3 until convergence of the SSE loss function

Functional Constraints in Parafac

Parafac Model: $\mathbf{X}_k = \mathbf{A}\mathbf{C}_k\mathbf{B}' + \mathbf{E}_k$ for $k = 1, \dots, K$

Assume that Mode A is the functional mode:

$$\mathbf{A} = \begin{pmatrix} \eta_1(1) & \eta_2(1) & \cdots & \eta_R(1) \\ \eta_1(2) & \eta_2(2) & \cdots & \eta_R(2) \\ \vdots & \vdots & \ddots & \vdots \\ \eta_1(I) & \eta_2(I) & \cdots & \eta_R(I) \end{pmatrix} = \mathbf{F}\boldsymbol{\alpha}$$

where

- $\eta_r(i) = \sum_{\ell=1}^{\nu} f_{\ell}(i)\alpha_{\ell r}$ is the r -th component function
- $\mathbf{F} = \{f_{\ell}(i)\}_{I \times \nu}$ where $f_{\ell}(\cdot)$ is the ℓ -th basis function
- $\boldsymbol{\alpha} = \{\alpha_{\ell r}\}_{\nu \times R}$ is a matrix of basis function coefficients

Functional Constraints in Parafac2

Parafac2 Model: $\mathbf{X}_k = \mathbf{A}_k \mathbf{C}_k \mathbf{B}' + \mathbf{E}_k$ subject to $\mathbf{A}_k' \mathbf{A}_k = \mathbf{\Phi} \quad \forall k$

Assume that Mode A is the functional mode:

$$\mathbf{A}_k = \begin{pmatrix} \eta_{1k}(1) & \eta_{2k}(1) & \cdots & \eta_{Rk}(1) \\ \eta_{1k}(2) & \eta_{2k}(2) & \cdots & \eta_{Rk}(2) \\ \vdots & \vdots & \ddots & \vdots \\ \eta_{1k}(I_k) & \eta_{2k}(I_k) & \cdots & \eta_{Rk}(I_k) \end{pmatrix} = \mathbf{F}_k \boldsymbol{\alpha}_k$$

where

- $\eta_{rk}(\cdot)$ is the r -th component function for the k -th level of Mode C
- $\mathbf{F}_k = \{f_{\ell_k}(i)\}_{I_k \times \nu_k}$ and $\boldsymbol{\alpha}_k = \{\alpha_{\ell_k r}\}_{\nu_k \times R}$
- $\sum_{i=1}^{I_k} \eta_{rk}(i) \eta_{sk}(i) = \phi_{rs} \iff \boldsymbol{\alpha}_k' \mathbf{F}_k' \mathbf{F}_k \boldsymbol{\alpha}_k = \mathbf{\Phi}$ for all k

Implementing Functional Constraints

CP or Tucker (Alsberg & Kvalheim, 1993; Timmerman & Kiers, 2002)

- $\mathbf{F}\boldsymbol{\alpha} = \tilde{\mathbf{U}}\tilde{\boldsymbol{\alpha}}$ where $\mathbf{F} = \tilde{\mathbf{U}}\tilde{\mathbf{S}}\tilde{\mathbf{V}}'$ and $\tilde{\boldsymbol{\alpha}} = \tilde{\mathbf{S}}\tilde{\mathbf{V}}'\boldsymbol{\alpha}$
- $\min_{\tilde{\boldsymbol{\alpha}} \in \mathbb{R}^{\nu \times R}} \|\mathbf{X}_A - \tilde{\mathbf{U}}\tilde{\boldsymbol{\alpha}}\mathbf{Z}'_A\|_F^2 \iff \min_{\tilde{\boldsymbol{\alpha}} \in \mathbb{R}^{\nu \times R}} \|\tilde{\mathbf{X}}_A - \tilde{\boldsymbol{\alpha}}\mathbf{Z}'_A\|_F^2$
- $\tilde{\mathbf{X}}_A = [\tilde{\mathbf{X}}_1, \dots, \tilde{\mathbf{X}}_K]$ where $\tilde{\mathbf{X}}_k = \tilde{\mathbf{U}}'\mathbf{X}_k$ denotes the projected data
- Given optimal $\tilde{\boldsymbol{\alpha}}$, the optimal coefficients are $\boldsymbol{\alpha} = \tilde{\mathbf{V}}\tilde{\mathbf{S}}^{-1}\tilde{\boldsymbol{\alpha}}$

Parafac2 (Helwig, 2017)

- $\mathbf{F}_k\boldsymbol{\alpha}_k = \tilde{\mathbf{U}}_k\tilde{\boldsymbol{\alpha}}_k$ where $\mathbf{F}_k = \tilde{\mathbf{U}}_k\tilde{\mathbf{S}}_k\tilde{\mathbf{V}}'_k$ and $\tilde{\boldsymbol{\alpha}}_k = \tilde{\mathbf{S}}_k\tilde{\mathbf{V}}'_k\boldsymbol{\alpha}_k$
- $\tilde{\boldsymbol{\alpha}}_k = \mathbf{R}_k\mathbf{H}$ where $\mathbf{R}'_k\mathbf{R}_k = \mathbf{I}_R$ (note: $\mathbf{R}_k$ is $\nu_k \times R$)
- $\min_{\mathbf{R}_k \in \mathbb{O}^{\nu_k \times R}} \|\mathbf{X}_k - \tilde{\mathbf{U}}_k\mathbf{R}_k\mathbf{H}\mathbf{C}_k\mathbf{B}'\|_F^2 \iff$
 $\min_{\mathbf{R}_k \in \mathbb{O}^{\nu_k \times R}} \|\tilde{\mathbf{X}}_k - \mathbf{R}_k\mathbf{H}\mathbf{C}_k\mathbf{B}'\|_F^2$
- $\tilde{\mathbf{X}}_k = \tilde{\mathbf{U}}'_k\mathbf{X}_k$ denotes the projected data and $\mathbb{O}^{\nu_k \times R}$ is the set of all $\nu_k \times R$ matrices with orthonormal columns

Some Constraint Options

Replace step of ALS algorithm with constrained LS update

Assume that Mode A is the constrained mode:

- Orthogonality: $\mathbf{A}'\mathbf{A} = \mathbf{I}_R$
- Non-negativity: $a_{ir} \geq 0$
- Monotonicity: $a_{ir} \leq a_{jr}$ for $i \leq j$
- Unimodality: $a_{ir} \leq a_{jr} \geq a_{kr}$ for $i \leq j \leq k$
- Functional: $a_{ir} = f_r(i)$ where $f_r(\cdot)$ is r -th component function
- Structural: $\mathbf{A} = \mathbf{A} * \mathbf{S}$ where $\mathbf{S}$ is known binary matrix

See Helwig (2017) for details on functional and structural constraints

General Framework

Most applications of constraints³ in Parafac/Parafac2 require solving the constrained multivariate least squares (CMLS) problem

$$\begin{aligned} \min_{\beta_1, \dots, \beta_p} \quad & \|\mathbf{Y} - \sum_{j=1}^p \mathbf{x}_j (\mathbf{Z}_j \beta_j)'\|_F^2 \\ \text{subject to} \quad & \mathbf{A}'_{0j} \beta_j = \mathbf{b}_{0j} \quad \text{and} \quad \mathbf{A}'_{1j} \beta_j \geq \mathbf{b}_{1j} \end{aligned} \tag{1}$$

where

- $\mathbf{Y} = \{y_{ik}\}_{n \times m}$ and $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_p] = \{x_{ij}\}_{n \times p}$ are known
- $\mathbf{Z}_j = \{z_{k\ell(j)}\}_{m \times \nu_j}$ are known basis function matrices
- $\beta_j \in \mathbb{R}^{\nu_j}$ are the unknown coefficients (for $j = 1, \dots, p$)
- $\mathbf{A}_{0j}$ and $\mathbf{b}_{0j}$ specify the equality (E) constraints
- $\mathbf{A}_{1j}$ and $\mathbf{b}_{1j}$ specify the inequality (I) constraints

³Orthogonality constraints do not fit into this general framework.

Vectorization or Back-Fitting?

If $\mathbf{Z}_j = \mathbf{Z} \forall j$, could solve the problem using a vectorization approach:

$$\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}'\mathbf{Z}'\|_F^2 = \|\text{vec}(\mathbf{Y}') - (\mathbf{X} \otimes \mathbf{Z})\text{vec}(\boldsymbol{\beta})\|_F^2 \quad (2)$$

where $\boldsymbol{\beta} = [\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_p]$ and $\otimes$ denotes the Kronecker product.

Could also use a back-fitting approach, which iteratively minimizes

$$\|\mathbf{Y}'_j - \mathbf{Z}_j\boldsymbol{\beta}_j\mathbf{x}'_j\|_F^2 \quad (3)$$

where $\mathbf{Y}'_j = \mathbf{Y}' - \sum_{h \neq j} \mathbf{Z}_h\boldsymbol{\beta}_h\mathbf{x}'_h$ and $\mathbf{x}_j$ denotes the j -th column of $\mathbf{X}$.

- Same as minimizing $\|\tilde{\mathbf{y}}_j - \mathbf{Z}_j\boldsymbol{\beta}_j\|_F^2$ where $\tilde{\mathbf{y}}_j = \mathbf{Y}'_j\mathbf{x}_j/(\mathbf{x}'_j\mathbf{x}_j)$

Flexible CMLS Solvers in R

The **CMLS** R package (Helwig, 2018a) provides flexible functions for solving CMLS problems under this general framework.

- `mslei()` solves CMLS problems with E/I constraints
- `mslun()` solves CMLS problems with unimodality constraints
- `cmls()` solves CMLS problems with 24 different options

Using vectorization or back-fitting, CMLS problem reduces to a CLS problem, which can be solved via quadratic programming routines.

- **CMLS** R package solves (a ridge-stabilized) QP problem
- Internally uses **quadprog** R package (Turlach & Weingessel, 2013), which implements the dual method of Goldfarb & Idnani (1983)

CMLS Simulation: Design

Consider the multivariate linear model: $\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E}$

- $\mathbf{Y} = \{y_{ik}\}_{n \times m}$ and $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_p] = \{x_{ij}\}_{n \times p}$ are known
- Goal is to estimate $\mathbf{B} = \{b_{jk}\}_{p \times m}$ via (constrained) least squares

For the simulation study, the parameters were generated as...

- $n = 50$ observations, $p = 2$ variables, and $m = 20$ measurements
- Entries of $\mathbf{X}$ independently sampled from $N(0, 1)$
- Entries of $\mathbf{E}$ independently sampled from $N(0, \sigma^2)$ for $\sigma \in \{0.1, 1\}$
- $\mathbf{X}$ and $\mathbf{B}$ (see next slide) are fixed throughout simulation

CMLS Simulation: Data Generating Parameters

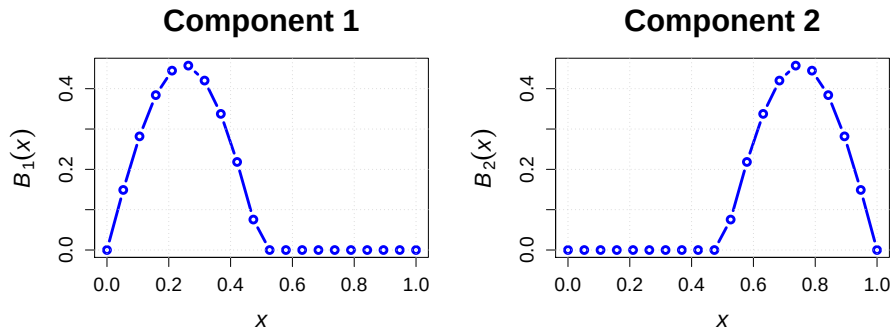


Figure 19: Rows of the data-generating $\mathbf{B}$ matrix used for the simulation study. Note that these parameters satisfy all constraints in Table 1 except the monotonicity (M) constraints.

CMLS Simulation: Analyses

For each level of σ , generate 1000 independent samples from the model.

For each generated sample, do the following:

1. Estimate $\mathbf{B}$ via `cmls()` using all 20 applicable constraints without including structural constraints
2. Repeat Step 1 including structural constraints that enforce $b_{1j} = 0$ for $j = 11, \dots, 20$ and $b_{2j} = 0$ for $j = 1, \dots, 10$

The above procedure was repeated using $\nu = 10$ and $\nu = 15$ degrees of freedom for the smoothness constraints.

CMLS Simulation: Results with $\sigma = 0.1$ and $\nu = 10$

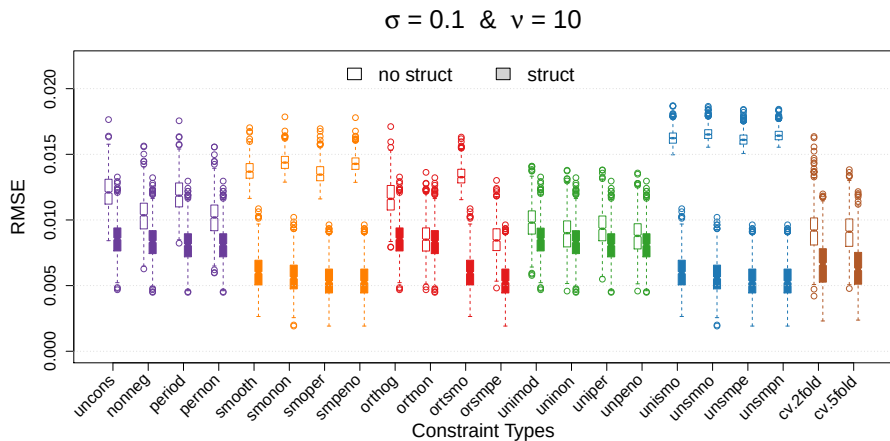


Figure 20: Simulation results (i.e., RMSE) for $\sigma = 0.1$ with $\nu = 10$.

CMLS Simulation: Results with $\sigma = 0.1$ and $\nu = 15$

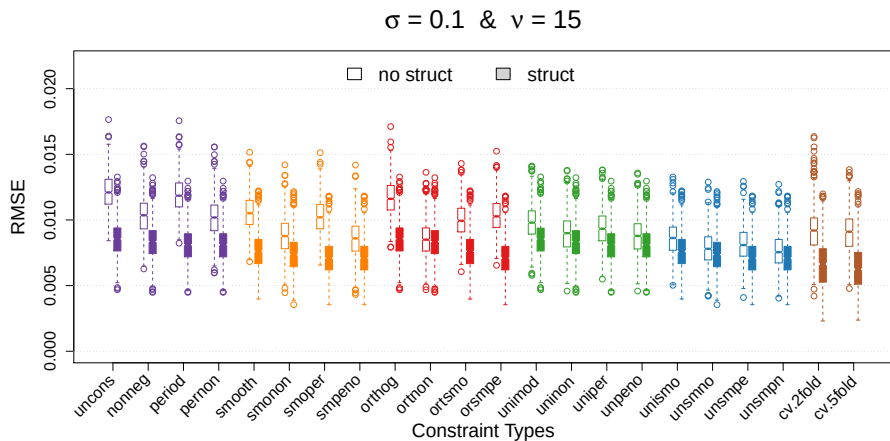


Figure 21: Simulation results (i.e., RMSE) for $\sigma = 0.1$ with $\nu = 15$.

CMLS Simulation: Results with $\sigma = 1$ and $\nu = 10$

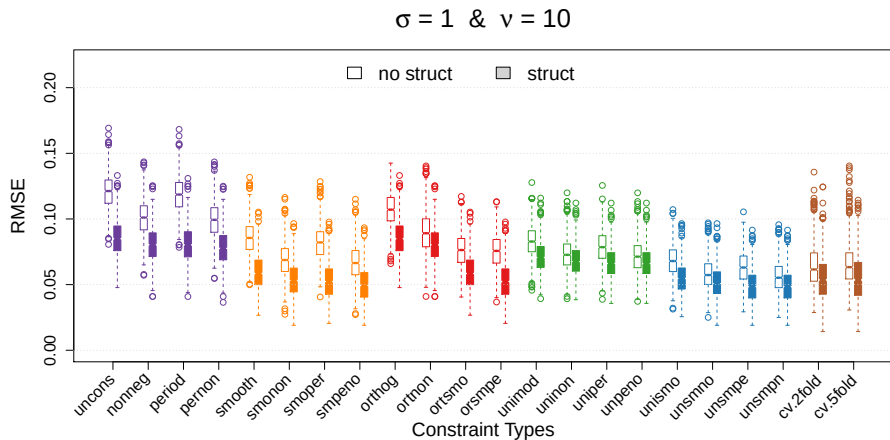


Figure 22: Simulation results (i.e., RMSE) for $\sigma = 1$ with $\nu = 10$.

CMLS Simulation: Results with $\sigma = 1$ and $\nu = 15$

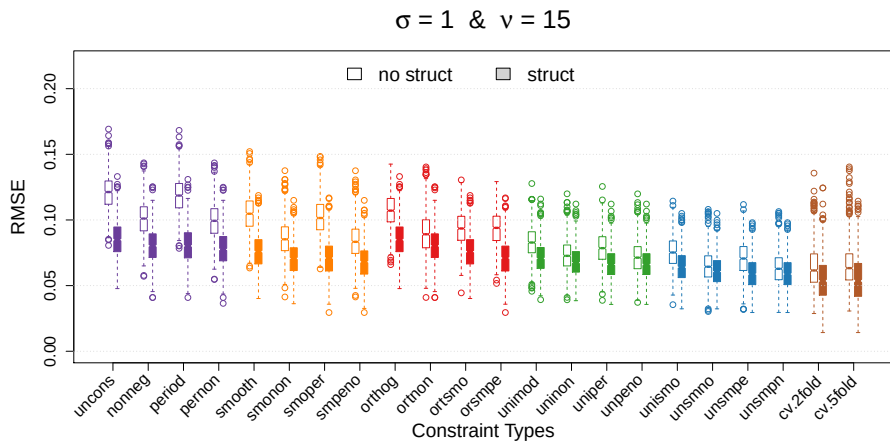


Figure 23: Simulation results (i.e., RMSE) for $\sigma = 1$ with $\nu = 15$.

CMLS Simulation: Results Summary

Structural constraints improve results for all constraint types.

Combined constraints generally out-performed single constraints.

Unimodality constraints did better than others—not surprising.

- Note that unimodality constraints include more information

Smoothness constraints improved the solution as long as the degrees of freedom parameter ν was set appropriately relative to σ .

- Cross-validation can be useful for tuning constraint options